Detailed Description of Data Analysis

Objective

This phase of data analysis is aimed at deriving insights from the fetched data by the APIs. It involves data exploration and interpretation emanating from multiple sources to derive insights into patterns, trends, and possible anomalies. Results are well-exposed in a visually appealing manner, as the easy comprehension of these figures is of utmost importance in making an informed decision based on the output from the application.

Steps of Data Analysis

1. Data Collection

It collects data from the following four APIs:

Virus Total: Threat intelligence on files, URLs, and domains by checking detection ratios and risk categories.

Abusive IPDB: Info on suspicious IP addresses, reports of abusive behavior like spamming or hacking.

IP Quality Score: Calculates fraud and phishing likelihood.

IP Info: Info about an IP, geolocation, ASN, and organization.

The data collected directly using APIs is a significant source of study. Each API provides various data points, which are then processed and analyzed for actionable insights.

2. Data Processing

The data is frequently presented in a different ordered format To synchronize data from several APIs, processing requires standardization factors such as IP address, threat type, and risk score.

Input	API Source	Risk Level	Geolocation	Final Verdict
12.33.55.76	AbuseIPDB, IPinfo	Low	Belleville, Michigan, US	Likely a safe site
https://mtnasmasklogiw.webflow.io	VirusTotal, Phishtank	Low	N/A	Likely a safe site
12.66.78.44	AbuseIPDB, IPinfo	Low	Piscataway, New Jersey, US	Likely a safe site
123.456.78.9	AbuseIPDB, IPinfo	Error	N/A	Likely a safe site
123.45.67.89	AbuseIPDB, IPinfo	Low	Seoul, South Korea	Likely a safe site
https://no.sototait.com/sbb/	VirusTotal, Phishtank	Low	N/A	Likely a safe site
https://mttamusklogiin.webflow.io	VirusTotal, Phishtank	Low	N/A	Likely a safe site
https://www.tinyurl.com/y7tm98kd	VirusTotal, Phishtank	Low	N/A	Likely a safe site
45.154.207.205	AbuseIPDB, IPinfo	High	Frankfurt am Main, Germany	Likely a phishing site
67.205.155.145	AbuseIPDB, IPinfo	High	North Bergen, New Jersey, US	Likely a phishing site
https://mylockers-points.com/	VirusTotal, Phishtank	Low	N/A	Likely a safe site
https://phantomwalletguide.webflow.io	VirusTotal, Phishtank	Low	N/A	Likely a safe site

3. Key Observations

3.1. High-Risk Inputs

Domains:

IP Quality Score classifies bside.com, bigstick.com, and applejuice.com as high risk with Fraud Scores ≥ 100

IPs:

165.227.147.218 (Frankfurt, Germany): Abuse Confidence 100, Reports: 6003.

13.248.169.48 (Seattle, US): Medium abuse (Abuse IPDB Confidence: 13), but flagged as phishing by IP Quality Score.

3.2. Low-Risk Inputs

google.com, kraken.com, apple.com (Virus Total and IPQualityScore findings: benign).

URLs with no matches in VirusTotal.

3.3 Geographic Trends

3.3. Failures

Input 123.456.78.9 fails AbuseIPDB and IPinfo analysis. However, its final verdict is still safe since there was insufficient evidence to corroborate it.

3.4. Geographic Trends

High-Risk Countries/Regions:

Frankfurt, Germany.

Seattle, Washington: Hosting high-risk domains on Amazon infrastructure.

Low-Risk Countries/Regions:

Chicago, Illinois, US (Google IPs).

Redondo Beach, California-safe infrastructure.

3.5 Recurring Threat Patterns

Domains like bigstick.com and biggie.com repeatedly classified as high-risk despite differing verdicts across APIs.

4. Analysis Insights

Insight 1: Geolocation Patterns

High-risk IPs cluster around known digital hubs, including Frankfurt and New Jersey. This may indicate hacked servers or targeted attacks against particular regions.

Insight 2: URL Safety

The URLs didn't exhibit phishing traits of current datasets, but more monitoring is advised.

Insight 3: API Errors

Errors in IP analyses have highlighted the need for fallback mechanisms to cross-check information with other APIs.

Insight 4: Abuse IPDB Reliability

Abuse confidence scores accurately identify malicious IPs, so it's a good source for finding phishing threats.

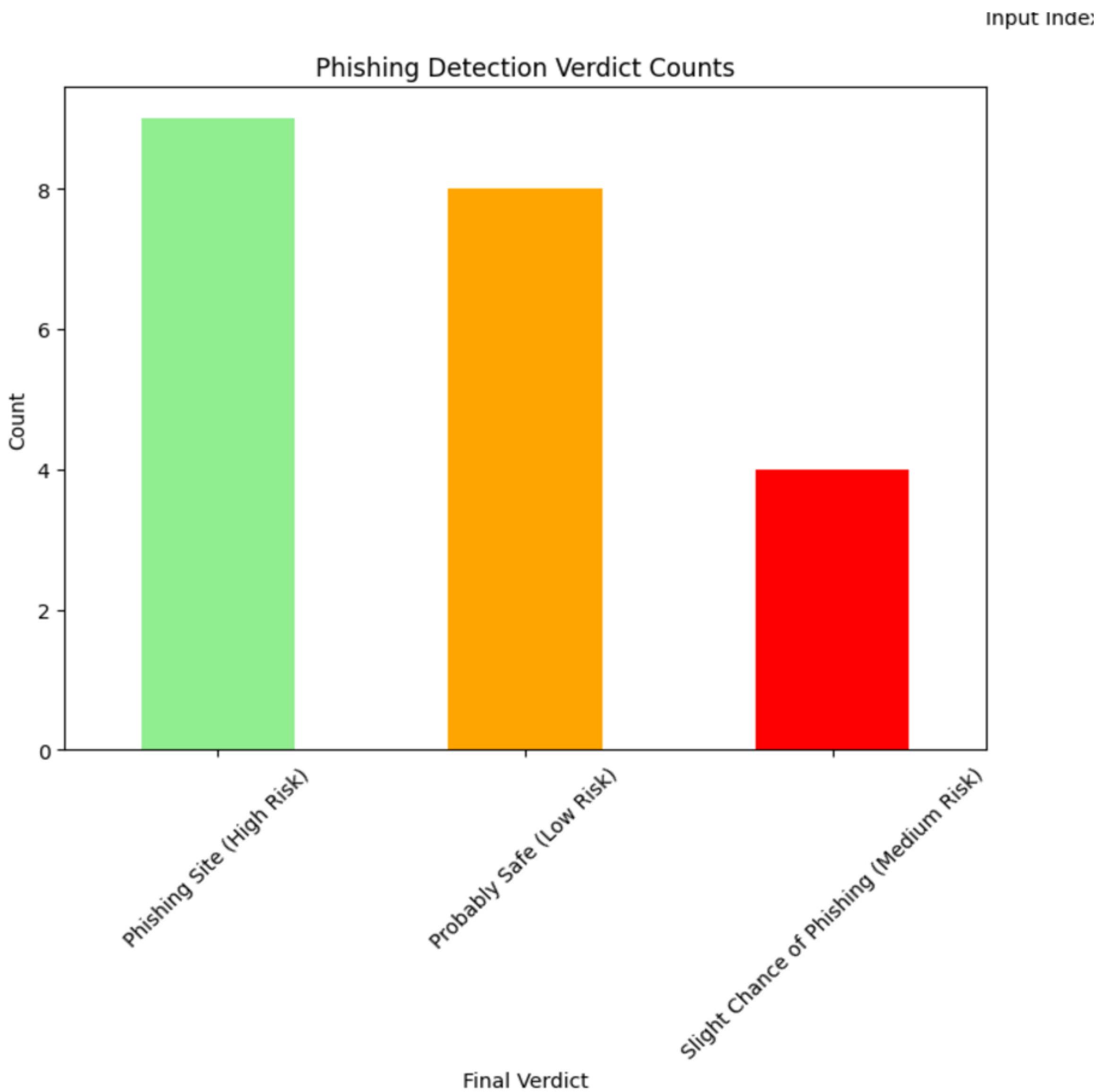
5. Visualization

5.1. Bar Chart of Threat Distribution

X-axis: Input types representing IP addresses and URLs.

Y-axis: threat levels, for example, Safe, Low-Risk, High-Risk

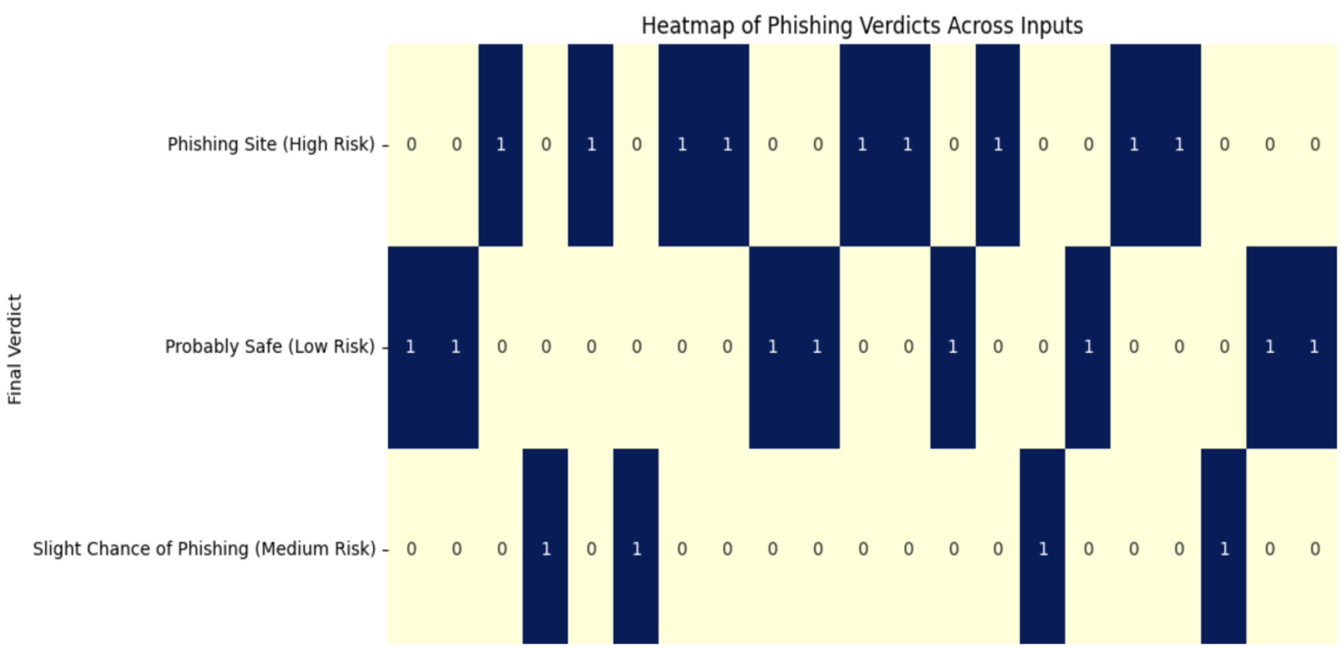
Insights: Most of the input entries are safe, but a small number of high-risk examples cause substantial outliers.



5.2 Heat Map: Geographic Distribution

Input: IP geolocation data

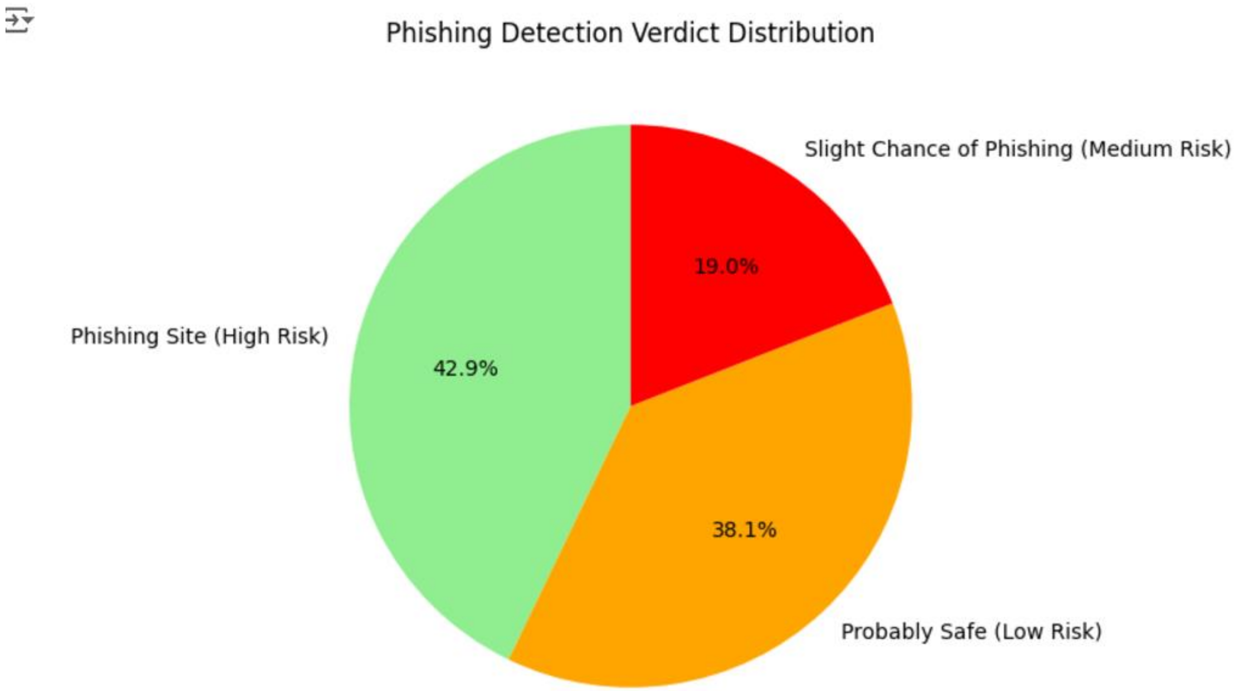
Insights: High-risk regions (e.g., Frankfurt, New Jersey) stand out.



5.3. Pie Chart: API Results

A breakdown of safe vs. flagged outcomes from each API:.

Insights: Some APIs, like Abuse IPDB, classify with more nuance than others.



6. Recommendations

6.1 Improved Threat Detection:

• Block high-risk domains, such as bigstick.com and biggie.com.

• Update the phishing datasets regularly with VirusTotal and PhishTank.

6.2 Geolocation-Based Alerts:

• Impose stricter regulations on inputs coming from high-risk locations such as Frankfurt and Seattle

6.3 Error Handling techniques:

• Add fallback techniques for API errors, including cross-validation with alternative sources

6.4 Continuous Dataset Changes

• Synchronize datasets with real-time changes from reliable sources like AbuseIPDB and VirusTotal

6.5 Utilize Advanced Analytics

• Employ clustering and anomaly detection to discern dynamic threats.

Tools and techniques:

Programming Language: Python; its libraries are pandas for data manipulation, matplotlib, and seaborn for data visualization.

APIs and Data Integration: To get this data, utilize Abuse IPDB, Virus Total, and IP info via requests for data fetching .

Visualization: Bar graphs, heat maps, and pie charts can represent the levels of threats, risk distributions geographically, or even breakdowns of API results.

Analysis: summarization of statistics, clustering, and anomaly detection to identify patterns- high-risk inputs.

Project Management: GitHub for source control and collaboration, Trello or Jira to monitor milestones and deliverables.